



UNIVERSIDAD
POLITÉCNICA
DE MADRID

Information Security

Parcial #3 – IWSIM21

29/04/2025



Surname:	Name:
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Part 1 of 2 (6 marks)

E-Corp uses digital signatures to guarantee the authenticity of emails sent by its executives. To achieve this, each executive has a digital certificate with their public key and corresponding private key. The digital certificates have been issued by the Certification Authority "E-Corp CA," and the email clients have been configured to accept certificates signed by it. To sign an email, it must attach the signature to the email along with the certificate used to sign it. The signing algorithm used for the emails is RSA with the *lenmod* hash function. This hash function consists of calculating the length of the message content (in characters, including spaces) modulo 91. Darlene Alderson, a member of the hacker group *fsociety*, plans to carry out an attack known as "CEO fraud" against E-Corp and its CEO, Philip Price. CEO fraud aims to trick employees with access to financial resources into paying a fake invoice or making a transfer from the company's account to a third party. Darlene has managed to forge the sender field of emails and proceeds to send the following email to the finance department:

From: PhilipPrice@ecorp.com
to: finance@ecorp.com
Subject: [URGENT] Transfer to supplier
Message:
"We must immediately transfer 20 million of USD to this account US33 2045 5787 3647 0001.
Philip Price
E-Corp CEO"
Message Length: 111

- 1- With this email, should the finance department employees proceed to make the transfer? Justify your response. [1 mark]

The email should be signed in order to accept its contents.

- 2- Darlene writes an email impersonating a journalist to CEO Philip Price, obtaining the following response:

From: PhilipPrice@ecorp.com
To: alexlevy@newsmorningshow.com
Subject: Re: Interview request
Message:
"I am sorry Alex but I am very busy these days and I cant participate in your program.
Philip Price
E-Corp CEO"
Message length: 108
Email signature: {75}
Certificate: Philip Price, Public key: {e=5,n=91}, Certificate Signature

- a) Verify Philip's signature [1,5 marks]

To verify the signature we first need to calculate the hash which results in 108. This mod 91 is 17. If we do $75^5 \bmod 91$ it should be 17.

- b) Would it be possible for Darlene to impersonate Philip Price to carry out the attack successfully? If the answer is yes, explain what Darlene should do to complete the attack step by step [3,5 marks]

Darlene needs to be able to sign certificates as Philip. To do this, she will have to find his private key. Given the key sizes used in this exercise, a simple factorization attack can be carried out to obtain the p and q corresponding to Philip Price's key. p and q are 13 and 7, respectively. From there, she can calculate $\phi(n)$, which is 72, and knowing the public key, she can find the modular inverse, which is 29 (this would be the private key) [1 point]. With that private key, she can already sign the message. First, she calculates the hash, which is $111 \bmod 91 = 20$ [0.5 points]. Now she can proceed to sign, which would be $20^{29} \bmod 91$, resulting in 76 [1 point]. Once she has the signature, she should send the same email but attach the obtained signature and Philip's certificate, which she extracted from his email [1 point].

Part 2 of 2 (3 marks)

The certificate chain on the next page has been returned when starting a connection to the website www.instagram.com. With the information provided about certificates answer the following questions:

- a) Write the name of the domain represented in the certificate returned by the website. [0,25 marks]

**.www.instagram.com*

- b) Write the start of the certificate's serial number. [0,25 marks]

01 AC 31 26 FF ...

- c) What company issued the certificate? [0,25 marks]

Digicert Inc

- d) Write the start of the serial number that should be used to verify the signature of the root certification authority. [0,25 marks]

02 AC 5C 26 6A...

- e) Explain briefly the purpose of the public key included in the website's certificate. [1,5 marks]

Encrypt the connection between the browser and the website with a symmetric encryption algorithm such as AES.

- f) The exponents of all three public keys displayed are all equal. Does this mean that all private keys are equal? Justify your answer [0,5 puntos]

The mod is different so the private keys will be different as well.

 DigiCert High Assurance EV Root CA

 DigiCert SHA2 High Assurance Server CA

 *.www.instagram.com



DigiCert High Assurance EV Root CA

Root certificate authority

Expires: Monday, 10 November 2031 at 01:00:00 Central European Standard Time

✔ This certificate is valid

> Trust

▼ Details

Subject Name

Country or RegionUS

OrganisationDigiCert Inc

Organisational Unitwww.digicert.com

Common NameDigiCert High Assurance EV Root CA

Issuer Name

Country or RegionUS

OrganisationDigiCert Inc

Organisational Unitwww.digicert.com

Common NameDigiCert High Assurance EV Root CA

Serial Number02 AC 5C 26 6A 0B 40 9B 8F 0B 79 F2 AE 46 25 77

Version3

Signature AlgorithmSHA-1 with RSA Encryption (1.2.840.113549.1.1.5)

ParametersNone

Not Valid BeforeFriday, 10 November 2006 at 01:00:00 Central European Standard Time

Not Valid AfterMonday, 10 November 2031 at 01:00:00 Central European Standard Time

Public Key Info

AlgorithmRSA Encryption (1.2.840.113549.1.1.1)

ParametersNone

Public Key256 bytes: C6 CC E5 73 E6 FB D4 BB ...

Exponent65537

Key Size2.048 bits

Key UsageVerify

Signature256 bytes: 1C 1A 06 97 DC D7 9C 9F ...



DigiCert SHA2 High Assurance Server CA

Intermediate certificate authority

Expires: Sunday, 22 October 2028 at 14:00:00 Central European Summer Time

✔ This certificate is valid

> Trust

▼ Details

Subject Name

Country or RegionUS

OrganisationDigiCert Inc

Organisational Unitwww.digicert.com

Common NameDigiCert SHA2 High Assurance Server CA

Issuer Name

Country or RegionUS

OrganisationDigiCert Inc

Organisational Unitwww.digicert.com

Common NameDigiCert High Assurance EV Root CA

Serial Number04 E1 E7 A4 DC 5C F2 F3 6D C0 2B 42 B8 5D 15 9F

Version3

Signature AlgorithmSHA-256 with RSA Encryption (1.2.840.113549.1.1.11)

ParametersNone

Not Valid BeforeTuesday, 22 October 2013 at 14:00:00 Central European Summer Time

Not Valid AfterSunday, 22 October 2028 at 14:00:00 Central European Summer Time

Public Key Info

AlgorithmRSA Encryption (1.2.840.113549.1.1.1)

ParametersNone

Public Key256 bytes: B6 E0 2F C2 24 06 C8 6D ...

Exponent65537

Key Size2.048 bits

Key UsageEncrypt, Verify, Derive

Signature256 bytes: 18 8A 95 89 03 E6 6D DF ...



*.www.instagram.com

Issued by: DigiCert SHA2 High Assurance Server CA

Expires: Tuesday, 6 May 2025 at 01:59:59 Central European Summer Time

✔ This certificate is valid

> Trust

▼ Details

Subject Name

Country or RegionUS

CountyCalifornia

LocalityMenlo Park

OrganisationMeta Platforms, Inc.

Common Name*.www.instagram.com

Issuer Name

Country or RegionUS

OrganisationDigiCert Inc

Organisational Unitwww.digicert.com

Common NameDigiCert SHA2 High Assurance Server CA

Serial Number05 28 A9 52 C8 C3 D1 2F C1 9B A3 7A 46 DC EF 90

Version3

Signature AlgorithmSHA-256 with RSA Encryption (1.2.840.113549.1.1.11)

ParametersNone

Not Valid BeforeTuesday, 4 February 2025 at 01:00:00 Central European Standard Time

Not Valid AfterTuesday, 6 May 2025 at 01:59:59 Central European Summer Time

Public Key Info

AlgorithmRSA Encryption (1.2.840.113549.1.1.1)

ParametersNone

Public Key256 bytes: 93 33 3D BA 71 CB 0A FA ...

Exponent65537

Key Size2.048 bits

Key UsageEncrypt, Verify, Wrap, Derive

Signature256 bytes: 0C A2 F3 3B 79 2B F0 20 ...